

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

- <think>
- First , convert 1 e V to joules . $1\text{ eV} = 1.602 \times 10^{-19}\text{ J}$.
 - Use the formula for kinetic energy : $\text{KE} = (p^2)/(2m)$, where p is momentum and m is mass .
 - The de Broglie wavelength is given by $\lambda = h/p$, where h is Planck's constant ($6.626 \times 10^{-34}\text{ Js}$).
 - Calculate the momentum p using $p = \sqrt{2m\text{KE}}$, where m is the mass of an electron ($9.109 \times 10^{-31}\text{ kg}$).
 - Substitute p into the de Broglie wavelength formula to find λ .
 - Convert the result to cm for easier comparison with the answer choices .

After calculations , the de Broglie wavelength is approximately $1.23 \times 10^{-7}\text{ cm}$.

</think>

<answer> $1.23 \times 10^{-7}\text{ cm}$ </answer>

